

Coding & Solution

Date	04 July 2026
Team ID	SOLO-CCFD-2026
Project Name	Credit Card Fraud Detection using Machine Learning
Maximum Marks	5 Marks

Solution Summary

	https://github.com/AyushaSanjuktha/credit-card-fraud-detection (update once pushed)
Programming Language(s)	Python
	Flask, scikit-learn, XGBoost, imbalanced-learn (SMOTE)
Key Features Implemented	Data preprocessing & SMOTE balancing; training/comparison of 4 classifiers; best-model selection by AUPRC; Flask web form for single-transaction prediction; CSV batch-prediction endpoint; IBM Watson ML deployment script
	Live IBM Watson ML endpoint not yet deployed with production credentials; authentication/authorization layer for analyst dashboard
Setup / Run Instructions	1) pip install -r requirements.txt 2) Place creditcard.csv in model/ and run train_model.py 3) cd flask_app && python app.py 4) Open http://127.0.0.1:5000

Code Quality Checklist

S.No	Criteria	Status (Yes/No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Partial
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes: Model selection prioritizes AUPRC and Recall over raw Accuracy, since fraud is under 0.2% of transactions and accuracy alone would be misleading.